Spherical Flow: Diffusivity Equation Derivation

Background: Building Blocks Used in the Derivation

Every diffusivity equation is built from the same three ingredients: a **mass balance** on a small control volume, **Darcy's law** for the flow rate, and **compressibility** to link density and porosity to pressure. Only the shape of the control volume (and so the flow area) changes between linear, radial and spherical systems. This section sets out the shared parts once.

B1. Assumptions and variables

Assumptions:

1. Single-phase, slightly compressible fluid
2. Homogeneous, isotropic rock: k constant. Fluid viscosity μ constant
3. Darcy's law applies (laminar flow)
4. No gravity effects
5. Isothermal
6. Small pressure gradients

Variables:

- ρ = fluid density
- v = Darcy velocity (volume flow rate per unit area)
- φ = porosity
- p = pressure
- c = fluid compressibility
- c_f = rock (pore) compressibility. The f stands for **formation** (the rock), not fluid. With several fluids, fluid compressibility carries a phase subscript (c_o, c_w, c_g). With a single fluid it is written simply as c.

B2. Mass crossing a face

Mass = density × volume. The volume of fluid crossing a face of area A at Darcy velocity v during time Δt is A·v·Δt. So

$$\text{mass} = \rho\, A\, v\, \Delta t$$

Units check: (kg/m³)·m²·(m/s)·s = kg ✓

B3. Darcy's law (differential form)

Flow goes from high to low pressure, at a rate proportional to the local pressure gradient:

$$v = -\frac{k}{\mu}\frac{\partial p}{\partial s}$$

where s is the direction of flow. For flow toward the center (the −r direction), s = −r, so ∂/∂s = −∂/∂r and

$$v = +\frac{k}{\mu}\frac{\partial p}{\partial r}$$

Pressure increases with r, so fluid moves inward from high to low pressure ✓. This **point form** holds in any geometry and at any time. The familiar q = kAΔp/(μL) is its integrated form for constant area and steady flow.

B4. Compressibility and the accumulation term

The mass stored in a control volume ΔV is the pore volume times the density. Fluid fills only the pore space, so the pore volume is φΔV, not ΔV:

$$\text{mass stored} = \rho\ (\ \phi\, \Delta V)\ = \Delta V\ (\ \phi\rho\ )$$

The porosity φ does not appear in the mass in/out terms because the Darcy velocity is already defined per unit total (bulk) area.

Both ρ and φ depend on pressure. **Fluid compressibility** and **rock compressibility** are

$$c = \frac{1}{\rho}\frac{\partial \rho}{\partial p}\ \Rightarrow\ \frac{\partial \rho}{\partial p} = c\,\rho, \qquad c_f = \frac{1}{\phi}\frac{\partial \phi}{\partial p}\ \Rightarrow\ \frac{\partial \phi}{\partial p} = c_f\,\phi$$

Since ρ and φ depend on position and time only through p, the chain rule gives

$$\frac{\partial \rho}{\partial s} = c\,\rho\,\frac{\partial p}{\partial s}, \qquad \frac{\partial \rho}{\partial t} = c\,\rho\,\frac{\partial p}{\partial t}, \qquad \frac{\partial \phi}{\partial t} = c_f\,\phi\,\frac{\partial p}{\partial t}$$

Apply the product rule to the accumulation term and substitute:

$$\frac{\partial\ (\ \phi\rho\ )}{\partial t} = \phi\frac{\partial \rho}{\partial t} + \rho\frac{\partial \phi}{\partial t} = \phi\, c\rho\frac{\partial p}{\partial t} + \rho\, c_f\phi\frac{\partial p}{\partial t}$$

Factor out φρ ∂p/∂t and define the **total compressibility** c_t = c + c_f:

$$\frac{\partial\ (\ \phi\rho\ )}{\partial t} = \phi\,\rho\,c_t\,\frac{\partial p}{\partial t} \qquad (\ B4\ )$$

For multiphase flow, c_t = c_f + S_o c_o + S_w c_w + S_g c_g. For one liquid phase it reduces to c + c_f.

B5. Why a small term can be dropped (linearization)

Expanding the flow term always produces a nonlinear piece of the form c (∂p/∂s)², multiplied by c. For a slightly compressible liquid, c is of order 10^{-6} to 10^{-5} psi^{-1} and pressure gradients are moderate away from the wellbore, so this term is far smaller than the second-derivative term. Dropping it gives a **linear** equation that can be solved analytically. The accuracy of this step is lowest for gases, where c varies strongly with pressure (pseudo-pressure is used instead).

B6. Hydraulic diffusivity

$$\eta = \frac{k}{\phi \, \mu \, c_t}$$

In SI units η has units of m²/s, the units of a diffusion coefficient. In oilfield units (k in md, μ in cp, c_t in psi^{-1}, t in hours):

$$\eta = 0.0002637 \, \frac{k}{\phi \, \mu \, c_t} \quad \left[\frac{\mathrm{ft}^2}{\mathrm{hr}}\right]$$

Spherical Flow Diffusivity Equation

When it occurs: fluid converges on a point from all directions. Examples are a well open over a very short interval (limited entry or partial penetration), and a formation-tester probe. Convention: flow toward the center, in the **−r direction**, with mass balanced over a time interval **Δt**.

Spherical Step 0: Geometry

Place a point sink at the origin. Around it, take a thin **spherical shell** between radius r and r + Δr.

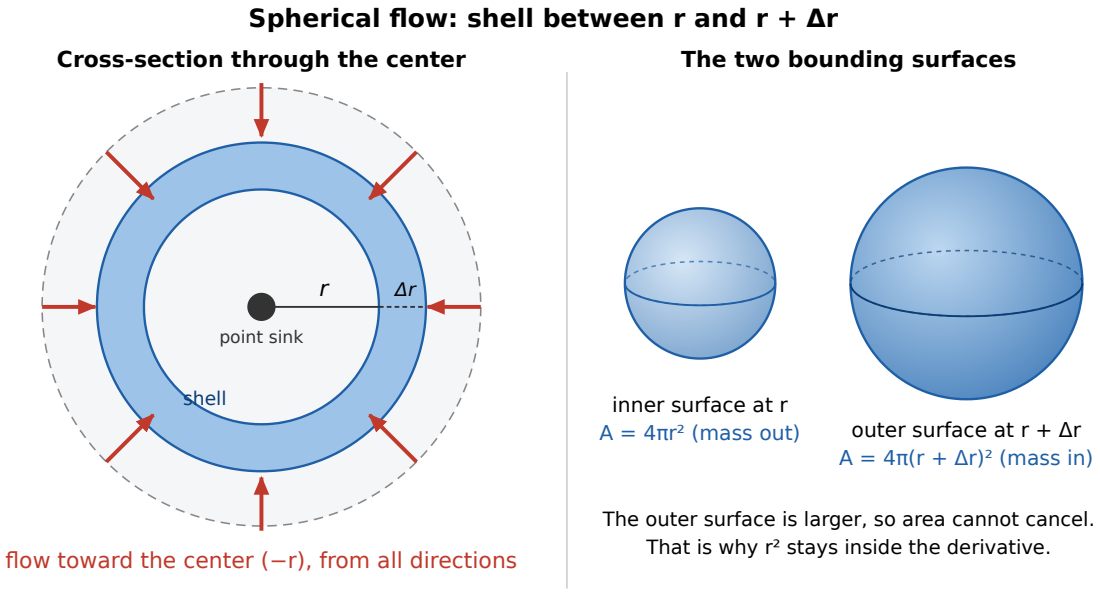


Figure 1. Spherical flow. Left: cross-section through the center showing the shell between r and r + Δr and flow converging on the point source. Right: the two spherical surfaces that bound the shell. The outer one has more area.

The flow area is the surface of a sphere, so it depends on r:

$$A\left(r\right) = 4\pi r^2$$

The volume of the shell is the difference of two spheres:

$$\Delta V = \frac{4}{3}\pi\left[\left(r+\Delta r\right)^3 - r^3\right] = 4\pi r^2\,\Delta r + 4\pi r\,\Delta r^2 + \frac{4}{3}\pi\,\Delta r^3$$

When Δr is very small, the terms in Δr² and Δr³ are negligible compared with the first:

$$\Delta V \approx 4\pi r^2\,\Delta r$$

Spherical Step 1: Mass crossing each face over Δt

Using mass = ρ A v Δt (Background B2). Fluid enters through the outer sphere at r + Δr and leaves through the inner sphere at r.

Mass in (outer face, at r + Δr):

$$\left(\text{ mass }\right)_{in} = 4\pi\,\Delta t\left[r^2\rho v\right]_{r+\Delta r}$$

Mass out (inner face, at r):

$$\left(\text{ mass }\right)_{out} = 4\pi\,\Delta t\left[r^2\rho v\right]_{r}$$

The factor r² stays inside the brackets because it is part of the area, and the area is different on the two faces. Only 4π comes out.

Spherical Step 2: Mass accumulated in the shell

Mass stored is ΔV(φρ) (Background B4), so

$$(\text{ mass })_{acc} = 4\pi r^2 \, \Delta r[\,(\,\phi\rho\,)_{\,t+\Delta t} - \,(\,\phi\rho\,)_{\,t}]$$

Spherical Step 3: Mass balance

(mass in) − (mass out) = (mass accumulated):

$$4\pi \, \Delta t(\,\left[r^2 \rho v\right]_{r+\Delta r} - \left[r^2 \rho v\right]_{r}) = 4\pi r^2 \, \Delta r[\,(\,\phi\rho\,)_{\,t+\Delta t} - \,(\,\phi\rho\,)_{\,t}]$$

Spherical Step 4: Divide and take limits

Divide both sides by 4π Δr Δt:

$$\frac{\left[r^2 \rho v\right]_{r+\Delta r} - \left[r^2 \rho v\right]_{r}}{\Delta r} = r^2 \, \frac{(\,\phi\rho\,)_{\,t+\Delta t} - \,(\,\phi\rho\,)_{\,t}}{\Delta t}$$

Take Δr → 0 and Δt → 0:

$$\frac{\partial\,(\,r^2 \rho v\,)}{\partial r} = r^2 \, \frac{\partial\,(\,\phi\rho\,)}{\partial t}$$

Divide by r² to get the **spherical continuity equation**:

$$\frac{1}{r^2}\frac{\partial\,(\,r^2 \rho v\,)}{\partial r} = \frac{\partial\,(\,\phi\rho\,)}{\partial t} \qquad (\,S1\,)$$

Spherical Step 5: Darcy's law

For flow toward the center (Background B3):

$$v = \,+\frac{k}{\mu}\frac{\partial p}{\partial r} \qquad (\,S2\,)$$

Spherical Step 6: Combine continuity and Darcy

Substitute (S2) into (S1) and pull the constants k and μ outside the derivative:

$$\frac{k}{\mu}\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \rho \frac{\partial p}{\partial r}\right) = \frac{\partial\,(\,\phi\rho\,)}{\partial t} \qquad (\,S3\,)$$

Spherical Step 7: Right-hand side (accumulation)

This is Background result (B4):

$$\frac{\partial\,(\,\phi\rho\,)}{\partial t} = \phi\rho\, c_t \frac{\partial p}{\partial t} \qquad (\,S4\,)$$

Spherical Step 8: Left-hand side (flow term)

The derivative is of a product of three factors: r², ρ and ∂p/∂r. The product rule gives:

$$\frac{\partial}{\partial r}\left(r^2\rho\frac{\partial p}{\partial r}\right) = 2r\rho\frac{\partial p}{\partial r} + r^2\frac{\partial p}{\partial r}\frac{\partial \rho}{\partial r} + r^2\rho\frac{\partial^2 p}{\partial r^2}$$

Use ∂ρ/∂r = c ρ ∂p/∂r:

$$= 2r\rho\frac{\partial p}{\partial r} + r^2\, c\rho\left(\frac{\partial p}{\partial r}\right)^2 + r^2\rho\frac{\partial^2 p}{\partial r^2}$$

Multiply by 1/r² and factor out ρ:

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\rho\frac{\partial p}{\partial r}\right) = \rho\left[\frac{\partial^2 p}{\partial r^2} + \frac{2}{r}\frac{\partial p}{\partial r} + c\left(\frac{\partial p}{\partial r}\right)^2\right] \qquad (\,S5\,)$$

Compare with radial flow, which had (1/r) ∂p/∂r. Here the coefficient is **2/r**, because the area grows as r² instead of r.

Spherical Step 9: Combine both sides

Substitute (S5) and (S4) into (S3), cancel ρ, and multiply by μ/k:

$$\frac{\partial^2 p}{\partial r^2} + \frac{2}{r}\frac{\partial p}{\partial r} + c\left(\frac{\partial p}{\partial r}\right)^2 = \frac{\phi\mu c_t}{k}\frac{\partial p}{\partial t} \qquad (\,S6\,)$$

Spherical Step 10: Linearization

Drop c (∂p/∂r)² (Background B5). The result is the **spherical diffusivity equation**:

$$\frac{\partial^2 p}{\partial r^2} + \frac{2}{r}\frac{\partial p}{\partial r} = \frac{1}{\eta}\frac{\partial p}{\partial t}$$

The left side equals (1/r²) ∂/∂r (r² ∂p/∂r). Check by the product rule: ∂/∂r(r² ∂p/∂r) = 2r ∂p/∂r + r² ∂²p/∂r², and dividing by r² gives back the two terms. The compact form is

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial p}{\partial r}\right) = \frac{1}{\eta}\frac{\partial p}{\partial t}, \qquad \eta = \frac{k}{\phi\mu c_t}$$

Spherical Step 11: A useful trick that reduces it to the linear equation

Define u = r p. Differentiate twice:

$$\frac{\partial u}{\partial r} = p + r\frac{\partial p}{\partial r}, \qquad \frac{\partial^2 u}{\partial r^2} = 2\frac{\partial p}{\partial r} + r\frac{\partial^2 p}{\partial r^2}$$

Divide the second equation by r:

$$\frac{1}{r}\frac{\partial^2 u}{\partial r^2} = \frac{\partial^2 p}{\partial r^2} + \frac{2}{r}\frac{\partial p}{\partial r}$$

This is exactly the left side of the spherical equation. Since ∂u/∂t = r ∂p/∂t, the equation becomes

$$\frac{\partial^2 u}{\partial r^2} = \frac{1}{\eta}\frac{\partial u}{\partial t}$$

So **u = r p obeys the 1D linear diffusivity equation**. This reduces the spherical problem to the linear one, which is why spherical flow problems have simple closed-form solutions.

Hemispherical flow

If the well is open only at the top or bottom boundary of the formation, flow converges on a half sphere. The flow area becomes $2\pi r^2$ instead of $4\pi r^2$. The diffusivity equation is unchanged, because the constant 2π divides out of both sides of the mass balance (just as 4π did). Only quantities that depend on the total flow area, such as the relation between rate and pressure, change.

Beyond Spherical Symmetry

If pressure also varies with the angles θ (from the vertical) and φ (around the axis), the same mass balance is done on an element with sides Δr, rΔθ and r sinθ Δφ. Its volume is $r^2 \sin\theta\, \Delta r \Delta\theta \Delta\varphi$. The three pairs of faces have areas

Pair	Face area
r-faces	$r^2 \sin\theta\, \Delta\theta\, \Delta\varphi$ (differs between r and r + Δr)
θ-faces	$r \sin\theta\, \Delta r\, \Delta\varphi$ (differs between θ and θ + Δθ)
φ-faces	$r\, \Delta r\, \Delta\theta$ (equal)

Whenever the area differs across the element, it stays inside the derivative. This is the same rule that kept r inside the derivative in radial flow. The result is the spherical Laplacian:

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial p}{\partial r}\right)+\frac{1}{r^2\,\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial p}{\partial\theta}\right)+\frac{1}{r^2\,\sin^2\theta}\frac{\partial^2 p}{\partial\phi^2}=\frac{1}{\eta}\frac{\partial p}{\partial t}$$

The $1/r^2$ and $1/(r^2\sin^2\theta)$ factors on the angular terms come from the arc-length factors in Darcy's law (distances $r\Delta\theta$ and $r\sin\theta\,\Delta\varphi$), plus one more $1/r$ from dividing by the element size, exactly as in the cylindrical θ-term. If pressure depends on r only, both angular terms vanish and the equation reduces to the spherical equation derived above. This last equation is stated here and follows the same steps as the cylindrical 3D derivation; it is not derived in full.

How the Three Systems Compare

System	Flow area A	Equation
Linear	constant	$\partial^2p/\partial x^2 = (1/\eta)\,\partial p/\partial t$
Radial (cylindrical)	$2\pi rh$	$\partial^2p/\partial r^2 + (1/r)\,\partial p/\partial r = (1/\eta)\,\partial p/\partial t$
Spherical	$4\pi r^2$	$\partial^2p/\partial r^2 + (2/r)\,\partial p/\partial r = (1/\eta)\,\partial p/\partial t$

For a general flow area A(s) along the flow path, all three are one equation: $(1/A)\,\partial/\partial s\,(A\,\partial p/\partial s) = (1/\eta)\,\partial p/\partial t$.